RESERVOIR

A community memory with consent built in

YEAR	ROLE	TEAM	STATUS
2026	Product, systems design, engineering	Solo build	Live

Summary

A free, open, community-grown memory of South African culture: plants, practices and languages, with community-controlled consent and a mission-native AI.

The problem

Cultural knowledge is often collected in a hurry and stored on someone else's terms. The people who hold it lose control of how it is attributed, reused and questioned.

Reservoir starts with consent as a data model, not a checkbox. Communities decide what can be shared, how it is credited and what the AI is allowed to do with it.

What is built

- Offline-first data capture with a Rust core API, a provider-agnostic Python AI layer and a mobile client.
- Knowledge sovereignty is modelled explicitly: communities control consent, attribution and reuse.
- Architecture decisions and safety notes are published in the repository.

Decisions that matter

Consent in the data model

Choice. Communities set attribution and reuse rules per record.

Why. Consent is enforceable, not a checkbox.

Trade-off. More states to model and explain.

Offline first

Choice. Capture works without a connection and syncs later.

Why. Contributors are not blocked by connectivity.

Trade-off. Sync and conflict handling add complexity.

Mission-native AI

Choice. The assistant serves the archive and its communities.

Why. The technology follows the mission, not the reverse.

Trade-off. A narrower assistant than a general chatbot.

Stack

`Rust` `Python` `TypeScript` `SQLite` `PostgreSQL`

Where to look

- Open live site: https://konethegreat.github.io/reservoir/
- Source on GitHub: https://github.com/konethegreat/reservoir
- Case study: konethegreat.github.io/work/reservoir